scientific reports

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A multimodal deep fusion framework for highway traffic anomaly detection

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This paper presents a novel framework for detecting and predicting abnormal traffic events on highways. Traditional traffic monitoring systems often rely on a single data source, which limits detection accuracy and robustness in complex environments. To address these challenges, we propose a multimodal deep fusion framework based on heterogeneous graph neural networks (HGNNs), enhanced by an ensemble contrastive pessimistic likelihood estimation (CPLE) algorithm. The framework integrates both static and dynamic traffic data, including video images, traffic flow, vehicle speed, and tunnel weather conditions. Through effective feature fusion, it significantly improves the accuracy and real-time performance of anomaly detection. Experimental results show that the model performs robustly across various scenarios, accurately identifying abnormal traffic events with high precision and stability. Compared with existing models such as AGC-LSTM and AttentionDeepST, the proposed MHGNN-CPLE model demonstrates superior performance, particularly in static detection tasks, achieving an accuracy of 0.980 and an F1 score of 0.967. In contrast, AGC-LSTM and AttentionDeepST achieve 0.965/0.945 and 0.960/0.935 in accuracy and F1 score, respectively. In dynamic scenarios, the model also maintains high accuracy under varying noise levels, indicating strong robustness. The research is motivated by the growing challenges of urbanization, where real-time detection and prediction of traffic anomalies are increasingly critical. Our framework effectively integrates multimodal data and leverages HGNNs to capture complex spatiotemporal dependencies, while the CPLE algorithm enhances robustness under uncertainty. The results confirm that the proposed method offers a reliable and accurate solution for real-time traffic anomaly detection, representing a significant advancement in intelligent transportation systems.

Keywords Multimodal deep fusion, Heterogeneous graph neural network, Traffic anomaly event, Dynamic detection, Static detection

As urbanization continues to accelerate, the management of highway traffic faces increasingly formidable challenges, particularly in the real-time detection and prediction of abnormal traffic events. Incidents such as accidents, congestion, and weather-induced disruptions significantly affect the efficiency and safety of road systems. Therefore, enhancing the accuracy and timeliness of anomaly detection is essential for reducing accidents, improving traffic flow, and ensuring public safety[1]. Most existing traffic monitoring systems rely on a single data source, such as surveillance camera, radar, or sensors, overlooking the inherently multimodal nature of traffic events. Consequently, traditional methods often lack sufficient accuracy and robustness in complex environments[2,3]. To address these limitations, this paper proposes a novel framework based on multimodal deep fusion and heterogeneous graph neural networks (HGNNs). By integrating diverse data sources, including images, point clouds, sound, and temperature, the framework captures the complex spatiotemporal dependencies present in traffic systems[4]. HGNNs are particularly well-suited for handling heterogeneous data and enable effective fusion of multimodal information, thereby improving the accuracy and reliability of abnormal event prediction[5].

Furthermore, to enhance the model's predictive performance, the Contrastive Pessimistic Likelihood Estimation (CPLE) algorithm is incorporated into the framework. This algorithm improves model robustness by optimizing prediction confidence through weighted likelihood adjustment[6]. In recent years, numerous studies

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have explored the application of deep learning and other intelligent algorithms to traffic anomaly detection. For instance, Liang et al.[7] developed a method for detecting abnormal sounds using a 1-D convolutional neural network (CNN) with a multi-head self-attention mechanism, enabling timely identification of events based on sound data. Zou et al.[8] applied the TL-XGBoost algorithm to detect anomalies in highway ETC transaction data, while Zhang et al.[9] proposed an anomaly detection method for retinal fundus images based on gated high-speed connections, which also holds potential for traffic monitoring. These works reflect growing interest in intelligent traffic monitoring; however, they primarily focus on single-modal detection and do not address the role of multimodal fusion in improving robustness.

The detection of traffic flow anomalies remains highly complex, often influenced by factors such as weather, road surface conditions, and equipment performance[10]. For example, Khan and Ahmed[11] presented a deep learning approach to predict accident severity under adverse weather, and Wang et al.[12] predicted highway icing using road surface temperature sensors. These studies highlight the potential of combining multimodal data with deep learning techniques for more effective anomaly detection. In this context, we present a multimodal anomaly detection framework built upon HGNNs, aiming to enhance the accuracy, reliability, and applicability of intelligent traffic systems in real-world environments.

Related work

Recent advances in traffic anomaly detection span a wide spectrum of methodologies, from rule-based systems to deep learning approaches. Traditional techniques often rely on traffic flow metrics, vehicle speed, or visual data from surveillance cameras, and employ statistical models or handcrafted rules to detect abnormalities. However, these systems struggle under complex, dynamic conditions and are highly sensitive to noise and environmental variability[13].

Machine learning methods have improved anomaly detection but often face challenges integrating heterogeneous data types[14]. For example, Han et al.[15] proposed a neural network–based framework for data cleaning on highway pavements, enhancing model accuracy by reducing noise but lacking adaptability to heterogeneous sources. Deep learning techniques, such as autoencoders[16] or pattern recognition models[17], have shown better performance in image-based anomaly detection but require extensive training data and are computationally intensive[18].

Graph Neural Networks (GNNs) have recently gained traction in traffic applications due to their ability to model spatiotemporal dependencies[19]. Heterogeneous GNNs (HGNNs), in particular, can effectively handle diverse node and edge types, making them suitable for multimodal data fusion[20]. Niu et al.[21] and Zhang et al.[22] explored GNNs for flow prediction and network security, while Zhou et al.[23] focused on visual anomaly trajectory detection. Yet, many such models remain limited to single-modality input or narrow scenarios like pavement cracks[24] or transactional ETC data[25,26].

This study extends existing GNN-based approaches by incorporating diverse sensor inputs—including images, point clouds, sound, and temperature—into a unified multimodal detection framework. Unlike prior methods, which often neglect sensor interdependencies or contextual correlations, our HGNN-based model captures multimodal interactions to improve robustness and generalization under real-world traffic conditions.

System design and model construction

The proposed model comprises three main components: a multimodal sensor preprocessing module, a heterogeneous graph neural network (HGNN) for spatiotemporal feature representation, and a contrastive pessimistic likelihood estimation (CPLE) strategy to enhance robustness under uncertainty. The overall architecture of the proposed system is illustrated in (Fig. 1).

Design of multimodal data acquisition and preprocessing methods

The essence of expressway abnormal events (such as accidents and congestion) is the result of multi factor coupling. Single modal data can only reflect local characteristics, while multi-modal data can provide complementary information through spatio-temporal alignment. In this study, four key types of multimodal data are collected: images, point clouds, sound, and temperature data. These data sources provide complementary information that helps the model effectively detect and predict abnormal traffic events. In particular, while image and point cloud data offer crucial visual and spatial information about traffic conditions, sound and temperature data provide valuable environmental context that improves the robustness and accuracy of the anomaly detection system.

Image data is obtained in real-time through highway cameras. Advanced computer vision algorithms, particularly convolutional neural networks (CNN) and region-based convolutional neural networks (R-CNN), are utilized to identify key elements within traffic scenes. In the preprocessing stage of image data, steps such as denoising (using techniques like Gaussian filtering or median filtering), scale normalization, and standardization are carried out. Additionally, components like multi-head attention and weighted graph convolution networks are employed to enhance the model's performance in complex traffic scenarios.

Point cloud data, provided by LiDAR sensors, captures the three-dimensional spatial information of roads and vehicles. The key to data preprocessing is to align and fuse point cloud data from different perspectives into a consistent 3D model. This study adopts a point cloud registration method based on the Iterative Closest Point (ICP) algorithm. The matching and fusion process of point clouds can be represented by the following mathematical model:

$$P_{fused} = \sum_{i=1}^{n} P_i \cdot w_i \tag{1}$$

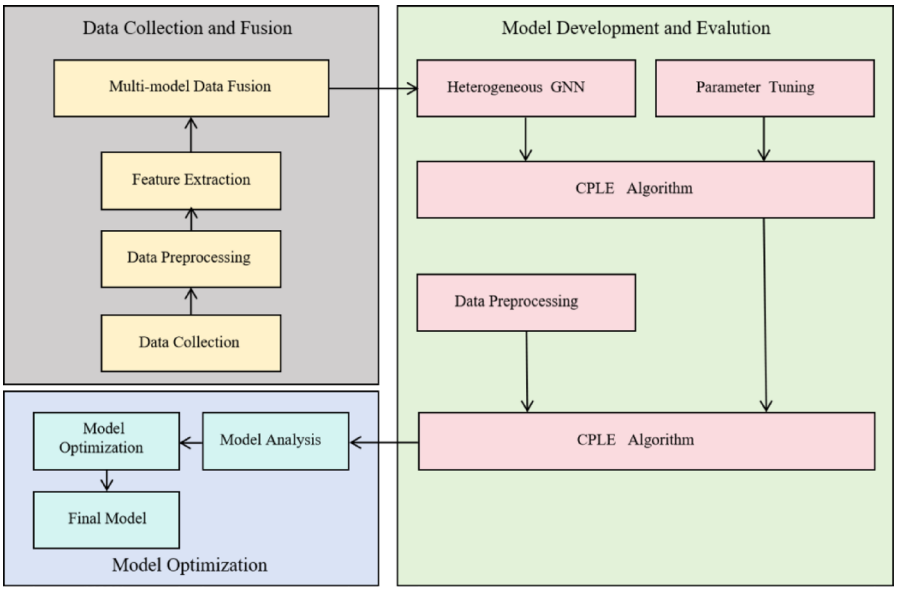


Fig. 1. Technical process of this article.

where P_i is the point cloud data collected by the i-th sensor, and w_i is the corresponding weight of each sensor. A fusion strategy based on distance and angle weighting is applied to enhance the spatial information expression ability of point cloud data.

Sound data, collected through sound sensors installed along highways, plays a crucial role in detecting traffic anomalies. Abnormal sounds, such as braking, engine noise, or collisions, can indicate incidents like accidents or disturbances that may not be visible through image data. The sound data preprocessing steps include background noise removal, frequency feature extraction, and Fourier transform. These steps help convert the sound signals into a frequency domain for further analysis. The preprocessing model for sound data can be represented as:

$$S_i^{freq} = FourierTransform(S_i) \tag{2}$$

$$S_i^{filtered} = Filter(S_i^{freq}, f_{threshold}) \tag{3}$$

$$S_i^{norm} = \frac{S_i^{filtered} - \mu_s}{\sigma_s} \tag{4}$$

where S_i is the original sound signal and S_i^{freq} is the transformed signal in the frequency domain, $f_{threshold}$ is the frequency threshold used in the filtering operation to retain only the relevant frequencies, μ_s and σ_s are the mean and standard deviation of the filtered sound data, and S_i^{norm} is the normalized sound signal. This ensures that the sound data is ready for subsequent fusion with other modalities.

Temperature data, collected through temperature sensors installed on the road surface, reflects environmental factors that can significantly affect traffic flow and safety. Temperature fluctuations, such as road icing during winter or extreme heat in summer, can lead to traffic disruptions, accidents, or congestion. Therefore, temperature data is an important modality for predicting weather-induced traffic anomalies. The preprocessing of temperature data involves interpolation to handle missing values and standardization to ensure consistency across different time periods and locations. The temperature preprocessing is modeled as follows:

$$T_i^{norm} = \frac{T_i - \mu_T}{\sigma_T} \tag{5}$$

where μ_T and σ_T are the mean and standard deviation of temperature data, and T_i^{norm} is the normalized temperature value.

After preprocessing, the data from different modalities are fused to create a comprehensive feature vector for anomaly detection. The fusion mechanism combines these heterogeneous data sources into a single vector that represents the traffic condition. The fusion process incorporates a weighted strategy where the contribution of each modality is dynamically adjusted based on its importance in detecting specific traffic anomalies. The fusion formula is given by:

$$x_{fused} = \sum_{i=1}^{n} w_i \cdot x_i^{norm} \tag{6}$$

where x_i^{norm} represents the normalized feature vector of each modality, and w_i is the weight of each modality. These weights are adjusted based on the historical performance of each modality and its relevance in

detecting different traffic events. For example, sound data may be more important in detecting accidents, while temperature data could be crucial for identifying weather-related anomalies. To present the logic of multimodal data acquisition and preprocessing more clearly, we write the algorithm as pseudocode shown in (Algorithm 1).

Heterogeneous graph neural network model design

Heterogeneous Graph Neural Networks (HGNNs) play a central role in the proposed framework for detecting and predicting abnormal traffic events on highways, as shown in (Fig. 2). This model design fully exploits the heterogeneous nature of various traffic data types and enables effective information fusion and representation learning through multi-level graph convolutional networks (GCNs). Traffic data exhibits strong spatiotemporal dependencies, and the relationships among different modalities are often complex and diverse. HGNNs facilitate efficient information propagation and learning by utilizing the unique structural characteristics of different node and edge types.

In this study, the nodes of the graph represent multimodal data sources, including traffic flow, vehicle speed, accident information, and environmental temperature, while the edges capture the interrelationships among

Input: Highway camera data, LiDAR point cloud data, road sound sensor data, road temperature sensor data

Output: Fused and preprocessed feature vectors

```
// Image data processing
image_data = GetRealTimeImageDataFromHighwayCameras()
denoised_image = DenoiseImage(image_data)
normalized_image = NormalizeScale(denoised_image)
enhanced_image_features = EnhanceFeatures(normalized_image)
// Point cloud data processing
point_cloud_data = GetPointCloudDataFromLiDAR()
aligned_point_clouds = AlignPointClouds(point_cloud_data)
fused_point_cloud = FusePointClouds(aligned_point_clouds)
// Sound data processing
sound_data = GetSoundDataFromRoadSensors()
cleaned_sound = RemoveBackgroundNoise(sound_data)
frequency_domain_signal = FourierTransform(cleaned_sound)
filtered_signal = FrequencyDomainFiltering(frequency_domain_signal)
// Temperature data processing
temperature_data = GetTemperatureDataFromRoadSensors()
interpolated_temperature = InterpolateMissingValues(temperature_data)
smoothed_temperature = SmoothData(interpolated_temperature)
denoised_temperature = Denoise(smoothed_temperature)
// Standardization and normalization of all data
standardized_image = Standardize(enhanced_image_features)
standardized_point_cloud = Standardize(fused_point_cloud)
standardized_sound = Standardize(filtered_signal)
standardized_temperature = Standardize(denoised_temperature)
// Weighted fusion of multimodal data
weights = CalculateWeightsBasedOnHistoricalData()
fused_feature_vectors = WeightedFusion([standardized_image, standardized_point_cloud,
standardized_sound, standardized_temperature], weights)
return fused_feature_vectors
```

Algorithm 1. Multimodal data acquisition and preprocessing.

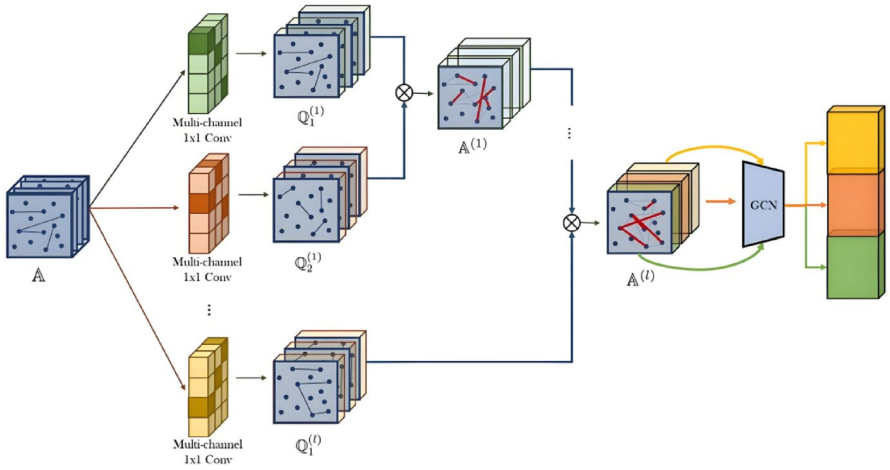


Fig. 2. Heterogeneous graph neural network structure.

these data types. A multi-level graph convolutional network architecture is designed to adaptively update node representations at each layer and effectively fuse features from different modalities. Node updates are performed through weighted aggregation, incorporating information from neighboring nodes and corresponding edge weights. The update process is formally defined by the following equation:

$$h_i^{(l+1)} = \sigma\left(\sum_{j\in N(i)} \frac{1}{c_{ij}} W^{(l)} h_j^{(l)} + b^{(l)}\right) \tag{7}$$

where $h_i^{(l+1)}$ represents the hidden state of the i-th node in the $(l+1)$-th layer, $N(i)$ denotes the set of neighboring nodes of the i-th node, $h_j^{(l)}$ is the hidden state of the j-th neighboring node in the l-th layer, $W^{(l)}$ is the weight matrix of the l-th layer, $b^{(l)}$ is the bias term, σ is the activation function, and c_{ij} is the edge normalization coefficient.

Moreover, to further enhance the model's capacity to perceive and distinguish heterogeneous data sources, Weighted Graph Convolutional Networks (Weighted GCNs) are employed. By introducing a weighting mechanism among different modalities, the model allows certain data sources to exert greater influence on the prediction results during training. The adaptive weighting strategy is formally represented by the following equation:

$$x_i^{(l)} = \sum_{j\in N(i)} \alpha_{ij} W^{(l)} x_j^{(l)} + b^{(l)} \tag{8}$$

where α_{ij} is the weighting coefficient between i and j nodes, which can be dynamically adjusted according to the strength of the relationship between nodes.

To fuse multimodal data, a weighted fusion strategy based on an attention mechanism is employed. This approach adaptively adjusts the contribution of each modality during the information propagation process. The weight assigned to each modality is computed according to the following equation:

$$\omega_i = \frac{exp(a_i)}{\sum_{i=1}^{n} exp(a_i)} \tag{9}$$

where a_i is the attention score representing the i-th modality, and ω_i denotes its corresponding weight. To further improve the prediction accuracy of the model, the Contrastive Pessimistic Likelihood Estimation (CPLE) algorithm is incorporated to optimize model parameters during training. This algorithm enhances the model's ability to recognize abnormal events by leveraging the relative differences between samples. The objective function of CPLE is defined as follows:

$$\mathcal{L}_{CPLE} = -\sum_{i=1}^{n} log\left(\frac{exp(z_i)}{\sum_{j=1}^{n} exp(z_j)}\right) \tag{10}$$

where z_i denotes the predicted output of the input data in the current model, and $\mathcal{L}_{CPLE}$ optimization objective aims to enhance the model's accuracy in detecting abnormal events by minimizing the loss function. The proposed heterogeneous graph neural network is designed to comprehensively capture the complex relationships among different modalities and to adaptively adjust the contributions through a combination of weighting and attention mechanisms. This design significantly improves the model's performance in predicting abnormal traffic events on highways. When integrated with the pessimistic likelihood estimation algorithm, the framework is capable

of accurately and efficiently identifying potential traffic anomalies in dynamic environments, offering timely alerts and decision support for traffic management systems. To clearly illustrate the architecture and logic of the proposed heterogeneous graph neural network, the corresponding algorithm is presented in pseudocode as presented in (Algorithm 2).

Integrated comparison pessimistic likelihood estimation (CPLE) algorithm design

The CPLE algorithm is particularly suitable for anomaly detection in highway traffic due to its ability to incorporate both prediction confidence and neighborhood consistency. By modeling anomaly scores using an exponentially decayed similarity function and adjusting prediction distributions accordingly, CPLE enhances the robustness of optimization under noisy or uncertain conditions. Furthermore, unlike common optimizers

Input: Multimodal data features, number of layers L
Output: Model predictions
// Initialize graph nodes and edges
1: nodes = InitializeNodes(multimodal_data_features)
2: edges = InitializeEdges(nodes)
// Initialize model parameters
3: for $l = 1$ to L do
4: $W^{(l)}$ = InitializeWeightMatrix()
5: $b^{(l)}$ = InitializeBiasTerm()
6: end for
// Forward propagation
7: for $l = 1$ to L do
8: for each node i in nodes do
9: $h_i^{(l+1)}$ = ActivationFunction(
10: Sum over j in $N(i)$ of $(1 / c_{ij}) * W^{(l)} * h_i^{(l+1)} + b^{(l)}$
11:)
12: end for
13: end for
// Weighted graph convolution
14: for each node i in nodes do
15: $x_i^{(l)}$ = Sum over j in $N(i)$ of $(\alpha_{ij} * W^{(l)} * x_j^{(l)} + b^{(l)})$
16: end for
// Multimodal data fusion with attention mechanism
17: attention_scores = CalculateAttentionScores(multimodal_data_features)
18: weights = CalculateWeightsBasedOnAttention(attention_scores)
19: fused_features = WeightedFusion(multimodal_data_features, weights)
// Combine with Contrastive Pessimistic Likelihood Estimation (CLL)
20: loss = CalculateCLLLoss(fused_features, true_labels)
21: UpdateModelParameters(loss)
22: predictions = MakePredictions(fused_features)
23: return predictions

Algorithm 2. Heterogeneous graph neural network model.

such as Adam or SGD, CPLE dynamically adjusts the learning rate and gradient magnitude based on sample difficulty and spatial influence, allowing the model to focus more on abnormal or ambiguous regions. This theoretically grounded adjustment mechanism aligns well with the multi-modal and heterogeneous nature of highway traffic data.

To further optimize the inference process of the CPLE algorithm, multiple mathematical models are introduced to jointly refine the sample weighting mechanism, loss function, and overall optimization strategy, aiming to improve the accuracy and robustness of anomaly event detection. In particular, the same weighting coefficients are dynamically adjusted based on similarity between samples.

The *i*-th sample's weighting coefficient is updated based on the difference between the predicted results z_i and those of neighboring samples z_j. The specific update formula is:

$$\omega_i^{(norm)} = \frac{\omega_i}{\sum_{i=1}^{n} \omega_i} \tag{11}$$

where *n* is the total number of samples, ω_i denotes the original weighting coefficient of each sample, and $\omega_i^{(norm)}$ represents the normalized weighting coefficient, ensuring that the total weight of across all samples sums to 1. On the basis of updating the weighting coefficients, the loss function in the CPLE algorithm is modified to account for the relative importance of each sample. Assuming the original loss function is $\mathcal{L}$, the weighted loss function is defined as follows:

$$\mathcal{L}_{weight} = \sum_{i=1}^{n} \omega_i^{(norm)} \cdot \mathcal{L}(y_i, \widehat{y}_i) \tag{12}$$

where $\mathcal{L}(y_i, \widehat{y}_i)$ is the standard loss function (such as cross-entropy loss), y_i is the true label of the *i*-th sample, $\widehat{y}_i$ is the predicted output of the model, and $\omega_i^{(norm)}$ is the normalized weighting coefficient corresponding to the *i*-th sample. To prevent the model from overfitting to abnormal samples and to ensure adequate learning from normal samples, a regularization term is incorporated into the loss function. This term penalizes excessive attention to outliers and balances the training process. L2 norm regularization or other suitable regularization techniques can be employed. The expression for the regularization term is given as follows:

$$\mathcal{L}_{reg} = \lambda \sum_{i=1}^{n} |\omega_i|^2 \tag{13}$$

where λ is the regularization coefficient, which controls the strength of the regularization term, ω_i denotes a model parameter, and $|\cdot|^2$ represents the L2 norm. During the optimization process, the CPLE algorithm measures the importance of abnormal samples by defining an anomaly score metric. Assuming the prediction error of the sample is $\epsilon_i = |z_i - y_i|$, the corresponding anomaly measure is computed as:

$$D_i = exp(-\alpha \cdot \epsilon_i) \tag{14}$$

where α is the adjustment parameter for anomaly measurement, and ϵ_i denotes the prediction error of the *i*-th sample. During the sample weighting process, the final prediction results of the model can be refined through weighted averaging, which enhances the model's sensitivity to abnormal samples. The weighted adjustment formula for the final prediction result is given as:

$$\widehat{y}_i^{(adjusted)} = \frac{\sum_{j=1}^{n} \omega_j^{(norm)} \cdot \widehat{y}_j}{\sum_{j=1}^{n} \omega_j^{(norm)}} \tag{15}$$

where $\omega_j^{(norm)}$ is the normalized weighting coefficient of the *j*-th sample, and $\widehat{y}_j$ denotes the predicted value of that sample. To further emphasize the influence of similar samples, a neighborhood influence metric is introduced. This metric adjusts weights by calculating the similarity between each sample and its surrounding samples. The calculation formula for the neighborhood influence is defined as:

$$N_i = \sum_{j=1}^{n} exp(-\beta \cdot |z_i - z_j|) \tag{16}$$

where z_j represents the predicted results of samples, and β is a hyperparameter that controls the sensitivity of the neighborhood influence measurement. To accelerate model convergence, the CPLE algorithm dynamically adjusts the learning rate based on sample weights. Assuming the current learning rate is η, the update formula for the dynamic learning rate is defined as follows:

$$\eta^{(t+1)} = \eta^{(t)} \cdot \left(1 + \gamma \cdot \sum_{i=1}^{n} \omega_i^{(norm)} \cdot \epsilon_i\right) \tag{17}$$

where $\eta^{(t)}$ is the current learning rate, γ is the learning rate adjustment rate, $\omega_i^{(norm)}$ denotes the normalized weighting coefficient of the i-th sample, and ϵ_i represents the prediction error of the sample. In order to optimize the weighted loss function, the gradient update during backpropagation is adjusted based on the sample weighting coefficients. Assuming the gradient of the loss function is ∇ L, the weighted gradient update is computed as:

$$\nabla \mathcal{L}_{\text{weighted}} = \sum_{i=1}^{n} \omega_i^{(norm)} \cdot \nabla \mathrm{L}(y_i, \widehat{y_i}) \tag{18}$$

where $\nabla \mathrm{L}(y_i, \widehat{y_i})$ denotes the standard gradient of the i-th sample, and $\omega_i^{(norm)}$ is its corresponding weighting coefficient. By incorporating these mechanisms, the CPLE algorithm enables the model to be more sensitive to abnormal event detection and adapt more flexibly to the characteristics and prediction difficulty of individual samples during training. To facilitate a clearer understanding of the Integrated Comparison Pessimistic Likelihood Estimation (CPLE) algorithm, its structure is presented in pseudocode format in (Algorithm 3).

Input: Training samples with predicted results z, true labels y, number of samples n, regularization coefficient λ, adjustment parameters α, β, learning rate η
Output: Optimized model
// Update sample weights
1: for i = 1 to n do
2: $\omega_i \leftarrow$ OriginalWeight(i)
3: $\omega_i^{(norm)} \leftarrow \omega_i$ / sum(ω_j for j = 1 to n)
4: end for
// Adjust loss function
5: Weighted_loss ← 0
6: for i = 1 to n do
7: $Loss_i \leftarrow$ StandardLossFunction(y_i, PredictedOutput_i)
8: Weighted_loss ← Weighted_loss + $\omega_i^{(norm)} * Loss_i$
9: end for
10: Regularization_term ← λ * sum(ω_i^2 for i = 1 to n)
11: Total_loss ← Weighted_loss + Regularization_term
// Calculate anomaly measure
12: for i = 1 to n do
13: $\epsilon_i \leftarrow |z_i - y_i|$
14: $D_i \leftarrow \exp(-\alpha * \epsilon_i)$
15: end for
// Adjust predictions
16: for i = 1 to n do
17: Adjusted_prediction_i ← sum($\omega_j^{(norm)}$ * PredictedOutput_j for j = 1 to n) / sum($\omega_j^{(norm)}$ for j = 1 to n)
18: end for
// Calculate neighborhood influence
19: for i = 1 to n do
20: $N_i \leftarrow$ sum($\exp(-\beta * |z_i - z_j|)$ for j = 1 to n)
21: end for
// Update learning rate
22: for t = 1 to number_of_training_epochs do
23: $\eta^{(t+1)} \leftarrow \eta^t * (1 + \gamma * \text{sum}(\omega_i^{(norm)} * \epsilon_i \text{ for } i = 1 \text{ to } n))$
24: end for

Algorithm 3. Integrated comparison pessimistic likelihood estimation (CPLE) algorithm.

Experiment and result analysis

Data set selection and experimental parameter setting and process

This study utilizes multiple real-world highway traffic datasets that encompass a wide range of traffic anomaly types, including car accidents, congestion, and weather-induced changes. These datasets, collected from diverse geographic regions and environmental conditions, provide rich feature representations and varied anomaly patterns, making them well-suited for analyzing traffic flow variations and predicting abnormal events on highways.

The primary highway traffic datasets used in this research include:

(1) METR-LA data set [traffic flow data of Los Angeles, https://github.com/liyaguang/DCRNN]: Originating from the highway network in the Los Angeles area, this dataset contains traffic flow data collected from multiple loop detectors. It includes key indicators such as vehicle speed, traffic density, and flow rate, recorded at five-minute intervals across various time periods. The dataset is well-suited for traffic flow prediction and abnormal event detection. It captures various traffic anomalies, including congestion, accidents, and road maintenance activities.
(2) PEMS-BAY data set [Bay Area traffic flow data, https://github.com/liyaguang/DCRNN]: Collected from the traffic monitoring system in the San Francisco Bay Area, this dataset contains extensive sensor data covering traffic flow and vehicle speed across multiple intersections. Data is recorded at five-minute intervals and includes key indicators such as vehicle speed and traffic volume. The dataset is suitable for traffic flow prediction, pattern analysis, and anomaly detection. It captures a variety of traffic anomalies, including weather-induced flow fluctuations, sudden accidents, and traffic control interventions.

Given the diverse origins and characteristics of these datasets, several preprocessing steps were implemented to ensure the quality and consistency of the model input data:

(1) Data Cleaning and Noise Removal: The datasets were initially cleaned to eliminate corrupted or missing entries, which are common in real-world traffic sensor data. Anomalous readings caused by sensor malfunctions were handled through outlier detection, followed by interpolation or exclusion as appropriate.
(2) Normalization and Standardization: Since the datasets originate from different regions (Los Angeles and the Bay Area) and may have varying value ranges (e.g., units of speed or density), all numerical features were normalized to a common scale. Min - max normalization was applied to features such as vehicle speed and traffic density to transform values into a standardized range (typically [0, 1]). This step helps prevent bias toward dominate features and improves training stability and convergence.
(3) Time Alignment: Given the shared temporal resolution of five-minute intervals, all data were synchronized to ensure temporal consistency across dataset. Time windows were adjusted where necessary to align observations, ensuring coherent representation of traffic patterns and anomaly events over time.
(4) Handling Missing Data: Missing data, an inherent challenge in traffic datasets, was addressed using interpolation techniques. Linear interpolation was applied to short-duration gaps, while more advanced methods such as forward or backward filling were employed for longer periods of missing data. These steps ensured continuity and reliability of the data for effective model training.
(5) Feature Engineering and Integration: To enhance model performance, additional features were derived by calculating rolling averages and temporal variations in traffic density across multiple time windows, thereby capturing short-term fluctuations. Data from different sources were then integrated into a unified format, where each data point included traffic attribute (e.g., flow, speed, density), along with corresponding timestamps and sensor location. This integration process aligned the METR-LA and PEMS-BAY datasets within a common feature space, enabling direct comparison and fusion.
(6) Anomaly Labeling: Traffic anomalies, such as congestion, accidents, and weather-related disruptions were labeled based on known event logs and data-driven detection patterns (e.g., sudden spikes or drops in traffic flow). This labeling enabled supervised learning, allowing the model to recognize and predict similar abnormal events with greater precision.

Through the aforementioned preprocessing steps, the input data were effectively standardized and cleaned, meeting the quality and consistency requirements necessary for accurate prediction and anomaly detection across different highway regions. The resulting integrated dataset contained uniform and relevant traffic features from multiple regions, enabling more robust analysis and reliable forecasting of traffic patterns and anomalies.

The training and validation sets were constructed to ensure that the model could effectively learn and generalize across various traffic scenarios. The training set consisted of traffic flow data from the METR-LA and PEMS-BAY datasets, which were normalized and time-aligned to include features such as vehicle speed, flow rate, and traffic density across multiple road segments and time periods. To address class imbalance, oversampling and undersampling methods were applied to balance normal and abnormal event samples. The validation set was randomly selected from the raw data to ensure no overlap with the training set, and was designed to evaluate the model's generalization performance under diverse traffic conditions. The training-to-validation split ratio was set at 80:20, ensuring sufficient learning while preserving a reliable evaluation set.

During the experiment, all datasets underwent a consistent preprocessing pipeline to ensure high data quality. This included filling missing values using interpolation and nearest neighbor method, detecting and correcting outliers through statistical methods, and standardizing all features to ensure compatibility with deep learning models. The data were segmented into time series according to experimental requirements to generate training and testing sets. A standard 80:20 split ratio was used between training and testing data to preserve both

data representativeness and the model's generalization capacity. The specific model parameter configurations are summarized in Table 1.

Comparison model and evaluation indicator selection

In the experimental simulation, in addition to implementing the proposed MHGNN-CPLE model, two advanced baseline models were selected for comparative analysis: the Adaptive Graph Convolutional Long Short Term Memory Network (AGC-LSTM) with a self-attention mechanism, and the Attention Deep Aggregation Model (AttentionDeepST). AGC-LSTM combines graph convolution networks with LSTM architectures and integrates a self-attention mechanism to effectively capture relevant spatiotemporal features during the fusion process, thereby enhancing prediction accuracy. The AttentionDeepST model employs a multi-level deep aggregation structure alongside a cross-mode self-attention mechanism, making it well-suited for complex fusion scenarios by improving both accuracy and stability. These models were selected to validate the effectiveness and advancement of the proposed algorithmic framework.

To comprehensively evaluate model performance, four widely used metrics were adopted: accuracy[27], recall[28], F1 score[29], and area under the curve (AUC)[30]. By comparing these indicators, we gain a multi-dimensional understanding of the model's capabilities relative to baseline methods in detecting abnormal traffic events. This provides a quantitative basis for evaluating the effectiveness of the proposed approach and guiding further model optimization.

Experimental results and performance analysis

The algorithm proposed in this study is applied to the detection and prediction of abnormal traffic events in both static and dynamic scenarios

As part of the analysis, this paper first examines a case involving abnormal traffic congestion caused by a tunnel traffic accident. Traffic flow information within the tunnel is collected using image acquisition sensors, as illustrated in Fig. 3. Initially, potential congestion risks in the monitored area are identified through basic vehicle detection and distance estimation techniques, enabling the early detection of abnormal traffic conditions.

From the detection results shown in the above figure, it can be observed that the traffic flow within the tunnel exhibits patterns based on a single video image. However, this information alone is insufficient to determine the specific type of abnormal event. To address this limitation, this study integrates additional modalities—namely, real-time average vehicle speed, traffic flow, and incident rate—corresponding to the time period. These features are fused to form a comprehensive representation that combines visual information, traffic environment data, and other contextual cues, as illustrated in (Fig. 4).

Based on the processing of the aforementioned feature data, the multimodal information was utilized to detect and predict abnormal events in highway traffic. The analysis was conducted across multiple scenarios, including normal conditions, extreme weather, traffic congestion, and accidents, as illustrated in (Fig. 5).

By predicting and analyzing traffic anomaly events under the three aforementioned scenarios, and integrating information such as weather conditions, traffic flow, and average vehicle speed across different road segments, the occurrence rates of various types of accidents were examined. The statistical results are presented in (Table 2).

Based on the experimental results presented above, the proposed model demonstrates the capability to detect and predict various types of traffic anomalies on highways using fused multimodal feature data. The results show that the model can effectively identify multiple categories of abnormal traffic events with high accuracy. In comparisons between actual event types and predicted results, the model achieves strong predictive consistency. For instance, in traffic congestion, the predicted event types closely align with actual observations, and the predicted anomaly probabilities are nearly identical to real-world values. For example, when the vehicle speed is 40.5 km/h and the traffic flow is 150.2 vehicles, the actual and predicted probabilities of congestion are 0.85 and 0.83, respectively. Similarly, for traffic accidents and weather-related anomalies, the model provides accurate predictions, with the differences between predicted and actual probabilities remaining within acceptable

Parameter	Set value	Explain
Figure neural network layers	3 layers	Set the number of layers in the graph neural network to extract spatial dependencies in the graph structure.
Learning rate	0.001	Initial learning rate, using Adam optimizer.
Batch size	64	The number of samples input during each training session.
CPLE algorithm sample weighting strategy	Linear weighting	The sample weighting strategy uses a linear weighting scheme to enhance the influence of different types of samples.
Time window length	10 time units (5 min/unit)	The sliding window length used for time series data.
Training epochs	100 rounds	The total number of rounds of network training until the model converges.
Model regularization parameters	0.0001	L2 regularization parameter used to prevent overfitting.
Activation function	ReLU	Activation function used for hidden layers to enhance non-linear fitting ability.
Hidden layer size	128 neurons	The number of neurons in the hidden layer of a graph neural network.
Dropout rate	0.5	Dropout ratio used to prevent overfitting.
Nonlinear effect modeling method	GCN combined with LSTM (dual-mode)	Combining graph convolutional networks (GCN) and long short term memory networks (LSTM) to model nonlinear interaction effects.
objective function	MSE (mean squared error)	Error measurement used for regression tasks to optimize the accuracy of model predictions.

Table 1. Experimental model parameter settings.

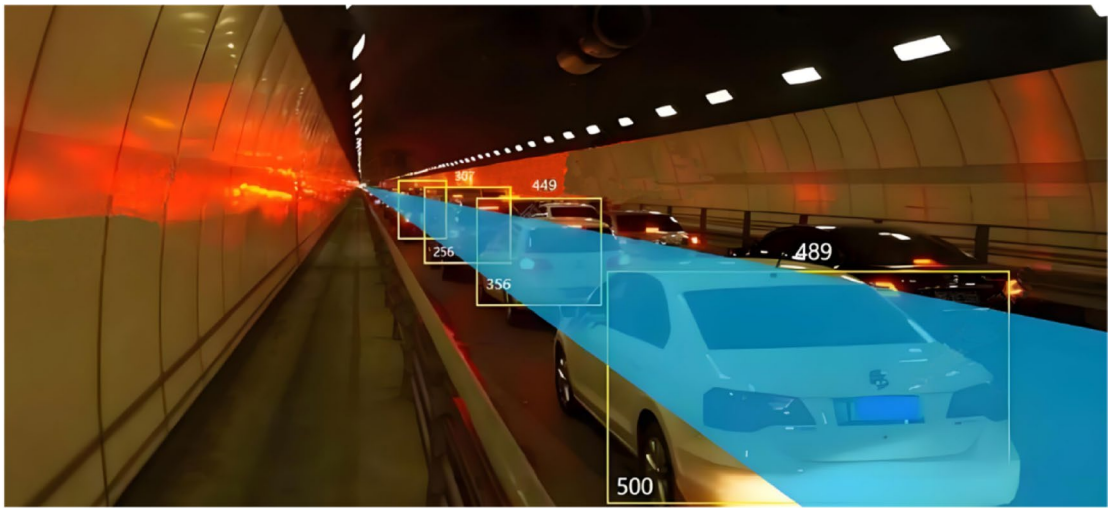


Fig. 3. Monitoring of image and video feature information under traffic congestion in tunnel.

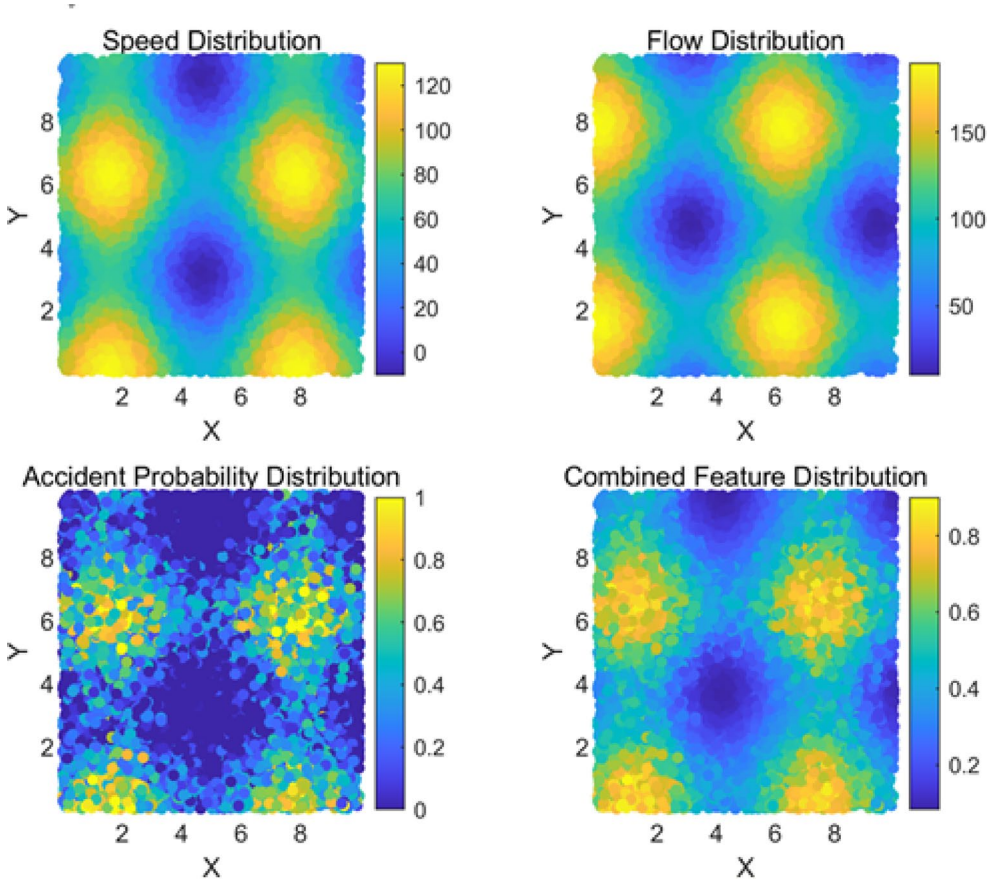


Fig. 4. Fused feature data.

margins —demonstrating strong recognition capabilities. Under normal traffic conditions, the model maintains low anomaly prediction probabilities, consistent with the actual situation. For example, at a vehicle speed of 60.4 km/h and a flow rate of 130.3 vehicles, the predicted and actual anomaly probabilities are 0.27 and 0.25, respectively. In summary, the integration of multimodal feature data significantly enhances the model's accuracy and stability in detecting and predicting traffic anomalies across various scenarios.

Furthermore, the detection results indicate that the proposed algorithm performs well in static anomaly analysis on highways. To further validate the dynamic detection capability of the framework, real-time video



Fig. 5. Typical prediction analysis scenario.

Speed (km/)	Flow (vehicles)	Weather (index)	Time of day (index)	Actual event type	Actual anomaly probability	Predicted event type	Predicted anomaly probability
40.5	150.2	0.80	0.9	Traffic congestion	0.85	Traffic congestion	0.83
20.3	90.7	0.90	0.2	Accident	0.92	Accident	0.91
60.4	130.3	0.75	0.6	Normal traffic	0.25	Normal traffic	0.27
35.8	175.0	0.70	0.8	Traffic congestion	0.78	Traffic congestion	0.79
50.2	100.1	0.85	0.4	Normal traffic	0.18	Normal traffic	0.20
5.0	50.0	1.00	0.3	Weather-related	0.90	Weather-related	0.88
15.2	95.3	0.95	0.1	Accident	0.91	Accident	0.89
55.0	120.0	0.70	0.7	Normal traffic	0.30	Normal traffic	0.32
80.3	160.2	0.60	0.5	Normal traffic	0.22	Normal traffic	0.23
10.5	180.1	0.85	0.9	Traffic congestion	0.80	Traffic congestion	0.82

Table 2. Prediction results of traffic abnormal events under different fusion information inputs.

data streams and sensor time-series information were incorporated to enable continuous monitoring of abnormal traffic events across highway sections. The perception results of input features across different scenario dimensions, along with the dynamically fused outputs, are illustrated in (Fig. 6).

As illustrated in Fig. 3, the results of pulse-based anomaly detection across four traffic event scenarios are summarized as follows:

In the "normal traffic" scenario, the signal remains smooth, with no significant fluctuation, and abnormal pulse activity is minimal. The peak fluctuation is approximately 0.3, indicating stable conditions.

In the "accident" scenario, a distinct pulse appears in the signal at 40-second mark, with an amplitude of 0.5. The detection algorithm successfully identifies and marks this anomaly, demonstrating strong sensitivity to sudden disruptions.

In the case of "traffic congestion", the signal exhibits periodic fluctuations with an amplitude of 0.3. Although the algorithm detects frequent abnormal pulses, the scenario shows a relatively high false alarm rate, indicating challenges in distinguishing between true congestion-related anomalies and normal fluctuations.

Under "weather impact" conditions, the signal displays a low-frequency cosine-like waveform with amplitude variations of approximately 0.2. While the algorithm accurately captures the anomaly trend, detection errors are slightly larger compared to other scenarios.

Furthermore, simulations experiments were conducted under varying noise levels (ranging from 0.05 to 0.3). The results show that as noise increases, the anomaly detection accuracy gradually declines—particularly in high-noise environments. However, the increase in detection error remains relatively small, highlighting the

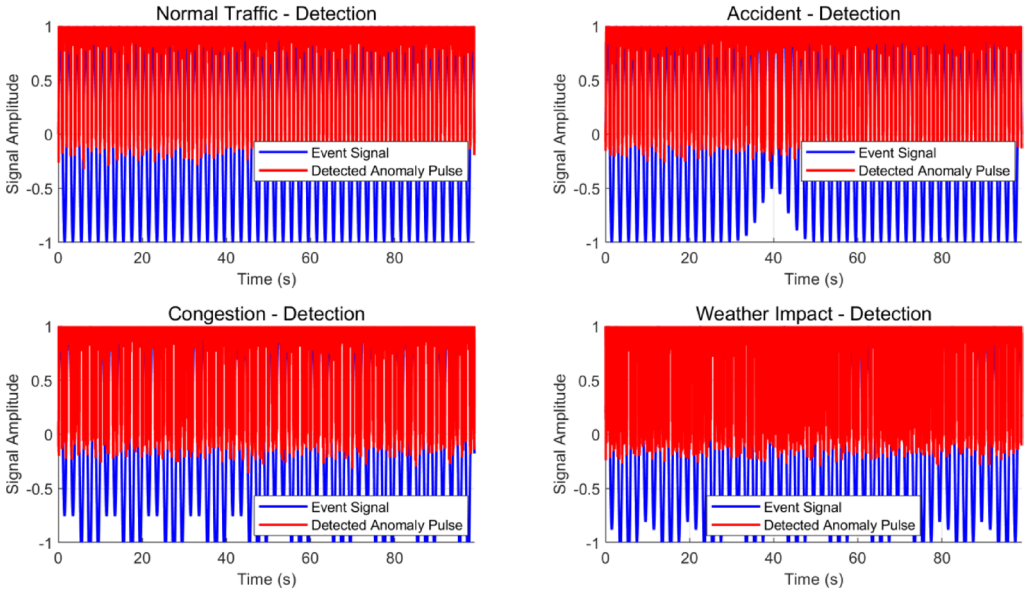


Fig. 6. Pulse signal diagram of dynamic abnormal event detection results in four different scenarios.

Event scenario	Accuracy	Precision	Recall rate	F1 score	AUC (area under curve)
Normal traffic	0.965	0.980	0.950	0.965	0.985
accident	0.960	0.980	0.940	0.960	0.975
traffic jam	0.950	0.965	0.930	0.960	0.970
Weather impact	0.955	0.975	0.930	0.952	0.980

Table 3. Overall performance indicators of the algorithm model in this Article for dynamic detection of traffic anomalies on highways.

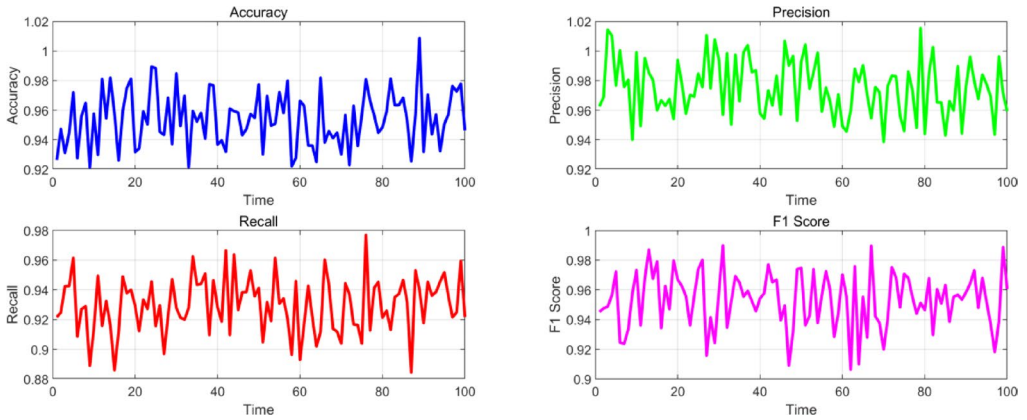


Fig. 7. Changes in various performance indicators in the time dimension under normal scenarios.

robustness of the proposed model to noise interference in real-time detection tasks. Table 3 shows the dynamic detection results across various evaluation metrics, averaged over 1,000 simulation trials.

To further assess the stability of various performance metrics across different traffic scenarios over time, this study monitored and analyzed the temporal evolution of key performance indicators during the experiment. The changes in performance parameters under each scenario were observed and recorded, as illustrated in (Figs. 7, 8, 9 and 10). The temporal variation of performance metrics under the normal traffic scenario is presented in (Fig. 7).

The changes in the time dimension of various indicators in the accident scenario are shown in (Fig. 8).

The changes in various indicators in the time dimension under congestion scenarios are shown in (Fig. 9).

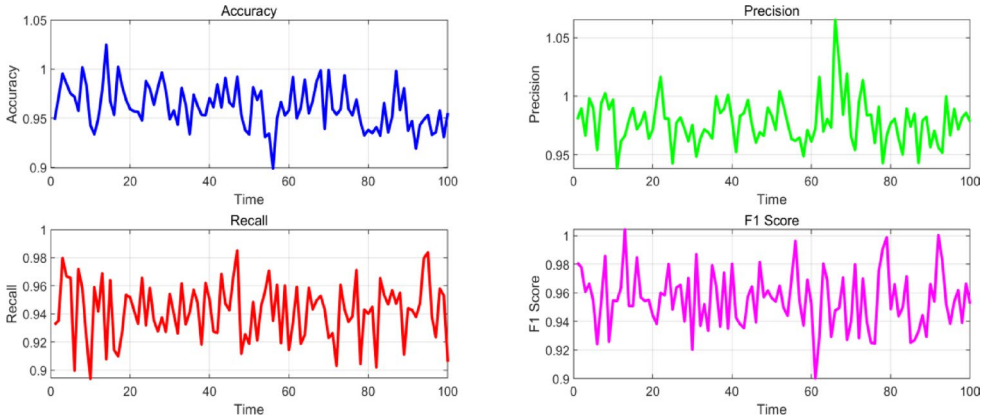


Fig. 8. Changes in various performance indicators in the time dimension under accident scenarios.

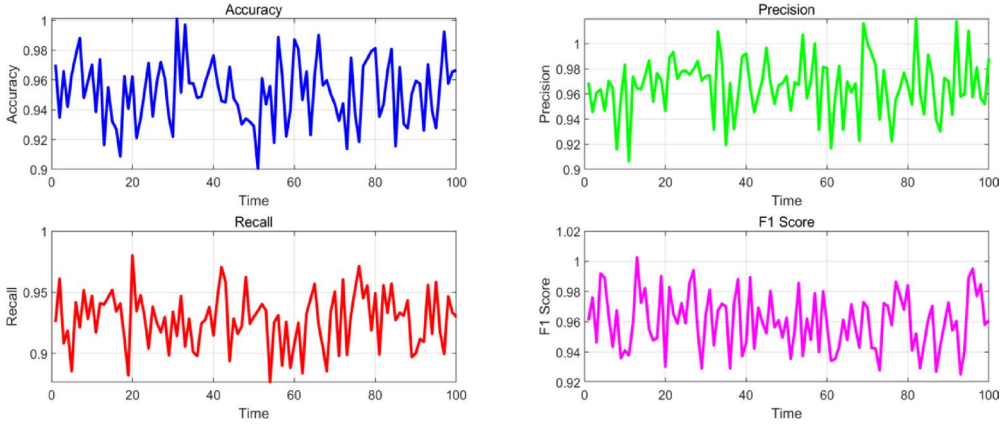


Fig. 9. Changes in various performance indicators in the time dimension under congested scenarios.

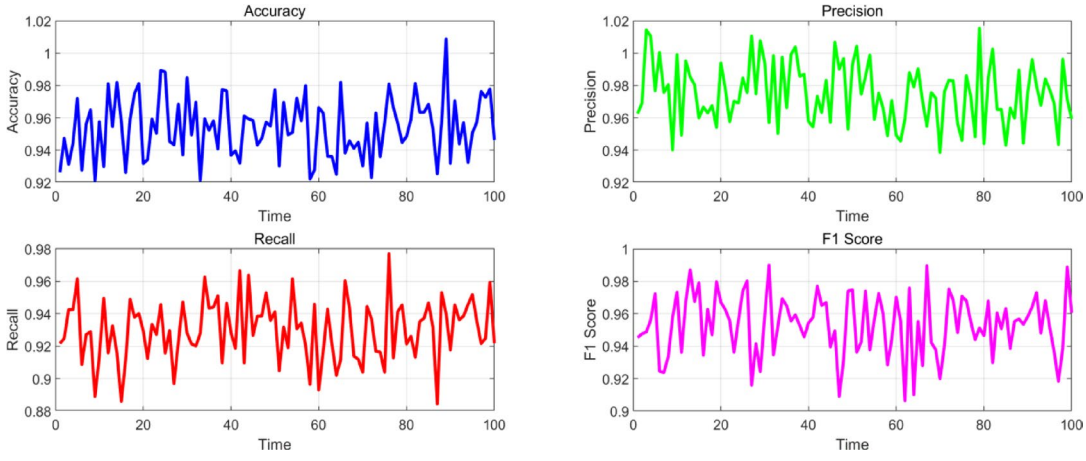


Fig. 10. Changes in various performance indicators in the time dimension under the influence of severe weather conditions.

The changing trends of various indicators in the time dimension under the influence of adverse weather conditions are shown in (Fig. 10).

Based on the experimental results, the proposed algorithm demonstrates strong overall performance in dynamically detecting traffic anomalies on highways. Across various event scenarios—including normal traffic, accidents, congestion, and weather-related disturbances—the model achieves high scores in key performance

Model algorithm	Accuracy	Precision	Recall rate	F1 score	AUC (area under curve)
MHGNN-CPLE	0.980	0.975	0.960	0.967	0.985
AGC-LSTM	0.965	0.950	0.940	0.945	0.960
AttentionDeepST	0.960	0.940	0.930	0.935	0.950

Table 4. Comparison of static detection performance (traffic flow scenario).

Model algorithm	Accuracy	Precision	Recall rate	F1 score	AUC (area under curve)
MHGNN-CPLE	0.975	0.960	0.950	0.955	0.980
AGC-LSTM	0.950	0.945	0.920	0.932	0.965
AttentionDeepST	0.940	0.930	0.910	0.920	0.960

Table 5. Comparison of static prediction performance (traffic flow prediction scenario).

metrics, such as accuracy, precision, recall, F1 score, and Area Under Curve (AUC). In the normal traffic scenarios, the model attains the accuracy of0.965, the precision of 0.980, the recall of 0.950, the F1 value of 0.965, and the AUC of 0.985, all of which indicate excellent detection capabilities. For accident and congestion scenarios, the performance is slightly lower but remains at a high level, with accuracy and F1 scores approaching 0.96. Under weather-influenced conditions, the model maintains relatively stable performance, with all evaluation metrics close to or exceeding 0.95. These results collectively demonstrate the robustness and reliability of the proposed algorithm across diverse traffic scenarios, confirming its effectiveness in dynamic anomaly traffic detection tasks.

Comparative experimental analysis with existing algorithm models

To validate the advancement of the proposed algorithm relative to existing models, this study conducts a comparative analysis during experimental simulations. Specifically, the designed MHGNN-CPLE model is evaluated against two baseline models that incorporate self-attention mechanisms: the Adaptive Graph Convolutional Long Short-Term Memory network (AGC-LSTM) and the Attention-based Deep Spatiotemporal Aggregation Model (AttentionDeepST).

As an initial focus, the static detection task under traffic flow scenario is considered. The task is designed to evaluate the models' abilities to accurately identify and classify traffic states (such as normal, congested, etc.) within specific time periods. Static detection typically requires high classification accuracy and reliability over time-invariant data segments. In this experiment, the performance of the MHGNN-CPLE, AGC-LSTM, and AttentionDeepST models is assessed and compared using standard evaluation metrics. The results of the static detection task are summarized in (Table 4).

The experimental results demonstrate that MHGNN-CPLE achieves superior performance in static detection tasks, with an accuracy of 0.980, significantly outperforming the other two models. The AGC-LSTM attains an accuracy of 0.950, but its relatively lower recall suggests the possibility of missed detections during the classification process. In contrast, although AttentionDeepST achieves a slightly higher accuracy (0.960) than AGC-LSTM, its lower precision and recall lead to a reduced F1 score of 0.935, indicating weaker overall classification performance. In summary, MHGNN-CPLE not only leads in accuracy but also achieves an AUC of 0.985, underscoring its robustness and reliability in processing traffic flow data under static conditions.

In static scenarios, traffic flow prediction relies on the model's ability to accurately capture temporal patterns from historical data. To evaluate predictive performance, this study further compares MHGNN-CPLE, AGC-LSTM, and AttentionDeepST in static traffic flow prediction tasks. The comparative results are presented in (Table 5).

As shown in the experimental results, MHGNN-CPLE achieves the best predictive performance among the evaluated models, with an accuracy of 0.975 and consistently high values in both precision and recall. These results highlight the model's strong capability in traffic flow prediction. Moreover, the AUC value of 0.980 further validates its stability and predictive reliability in real-world scenarios. In comparison, AGC-LSTM exhibits relatively balanced performance, with an accuracy of 0.950, but slightly lower values in precision and recall. Its F1 score of 0.932 suggests that it may adopt a more conservative strategy in traffic flow prediction, potentially leading to under-detection in some cases. AttentionDeepST, however, performs less effectively in static prediction tasks. A noticeable gap between its accuracy and recall results in a lower F1 score of 0.920, indicating weaker overall predictive performance compared to the other two models. The comparative results are visualized in (Fig. 11).

Secondly, in dynamic scenarios, the primary objective is to identify and detect traffic accidents in real time. The core challenge of dynamic detection lies in accurately capturing the precise moment an accident occurs and classifying it effectively. This task requires models to respond rapidly and reliably within continuously. To evaluate the dynamic detection capability of the proposed model, a comparative experiment was conducted using the MHGNN-CPLE, AGC-LSTM, and AttentionDeepST models. Their performance was assessed in terms of accuracy, recall rate, and F1 score within accident detection scenarios. The results of this evaluation are presented in (Table 6).

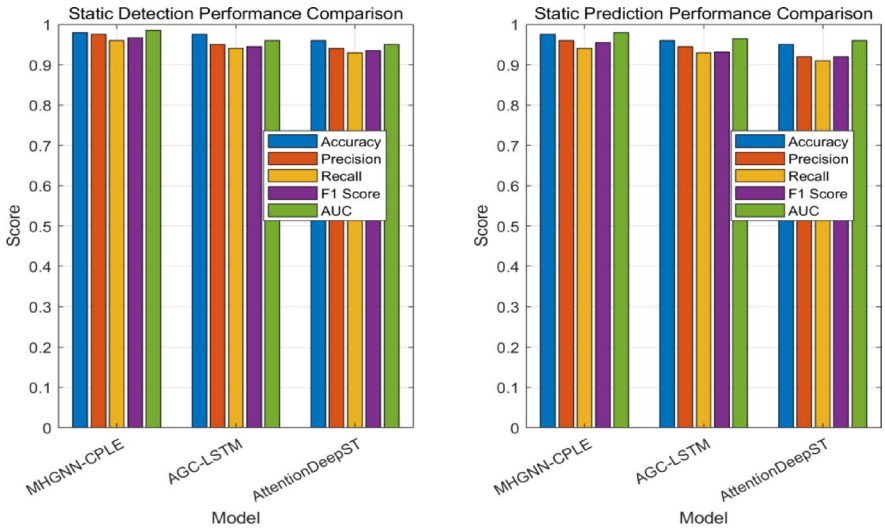


Fig. 11. Comparison of detection and prediction performance of different algorithm models in static scenes.

Model algorithm	Accuracy	Precision	Recall rate	F1 score	AUC (area under curve)
MHGNN-CPLE	0.950	0.960	0.930	0.945	0.970
AGC-LSTM	0.940	0.950	0.910	0.930	0.960
AttentionDeepST	0.920	0.920	0.890	0.905	0.950

Table 6. Comparison of dynamic detection performance (Accident detection Scenario).

Model algorithm	Accuracy	Precision	Recall rate	F1 score	AUC	Robustness	Significance	Stability
MHGNN-CPLE	0.940	0.950	0.920	0.935	0.965	High	Very significant	Very high
AGC-LSTM	0.925	0.930	0.900	0.915	0.950	Medium	Significant	High
AttentionDeepST	0.910	0.900	0.880	0.890	0.930	Medium	Moderate	Medium

Table 7. Comparison of dynamic prediction performance (accident prediction scenarios).

The experimental results indicate that MHGNN-CPLE continues to demonstrate its advantages in dynamic accident detection tasks. It achieves an accuracy of 0.950, precision of 0.960, recall of 0.930, F1 score of 0.945, AUC of 0.970, reflecting strong and balanced performance across all key indicators. In comparison, while AGC-LSTM also performs well, its recall rate is slightly lower (0.910), which may lead to missed detections in time-sensitive accident scenarios. AttentionDeepST, by contrast, shows relatively modest performance in this task, with an accuracy of 0.920 and a recall of 0.890. Its overall results are more conservative, potentially resulting in a higher rate of missed accident reports. These findings highlight the comprehensive strengths of MHGNN-CPLE, which consistently outperforms the baseline models in dynamic accident detection, particularly in balancing detection accuracy with real-time responsiveness.

Dynamic accident prediction plays an important role in intelligent traffic management, requiring models to analyze real-time traffic data and forecast potential accidents occurrences. The primary challenge of this task lies in the model's ability to effectively capture temporal dependence fluctuations in traffic flow patterns. In this experiment, the performance of three models—MHGNN-CPLE, AGC-LSTM, and AttentionDeepST—was evaluated in the context of dynamic accident prediction. The comparative results are presented in (Table 7).

In accident prediction scenarios, the MHGNN-CPLE model performs the best with the highest performance indicators, especially in terms of accuracy (0.940), precision (0.950), recall (0.920), and AUC (0.965), demonstrating excellent predictive ability. In addition, its robustness was rated as "high", significance as "very significant", and robustness as "very high", indicating that the model can maintain excellent stability and reliability in different data environments. In contrast, although the AGC-LSTM model performs slightly worse with an accuracy of 0.925, precision of 0.930, recall of 0.900, F1 score of 0.915, and AUC of 0.950, it still demonstrates strong performance. Its robustness is "moderate", significance is "significant", and robustness is "high", indicating that the model performs well in most cases, but may be slightly inadequate in certain specific contexts. The AttentionDeepST model ranks last among all indicators, with an accuracy of 0.910, precision of 0.900, recall of 0.880, F1 score of 0.890, and AUC of 0.930, indicating relatively low predictive performance. Its robustness is "moderate", significance is "moderate", and robustness is "moderate", indicating that the model may

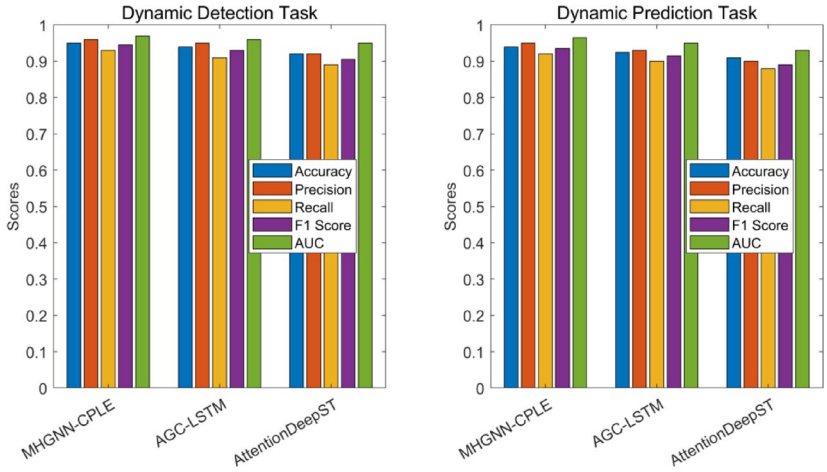


Fig. 12. Performance Comparison of Dynamic Detection and Prediction in Different Scenarios.

Model Algorithm	Accuracy	Precision	Recall	F1 score	AUC
MHGNN-CPLE	0.945	0.950	0.930	0.940	0.970
AGC-LSTM	0.920	0.930	0.900	0.915	0.955
AttentionDeepST	0.905	0.910	0.880	0.895	0.945

Table 8. Cross-Dataset generalization Performance.

not perform as well as the first two when dealing with complex and variable data. Overall, MHGNN-CPLE is the most competitive model among the three, suitable for application scenarios that require high accuracy and stability. AGC-LSTM is suitable for most applications, but may not perform as well as MHGNN-CPLE under extreme or complex conditions. AttentionDeepST is suitable for simpler application scenarios, where accuracy requirements are more relaxed. The comprehensive comparison is shown in Fig. 12.

The experimental results demonstrate that the proposed framework, which integrates multimodal deep fusion and heterogeneous graph neural networks (HGNNs), achieves superior accuracy and robustness in detecting and predicting abnormal traffic events compared to traditional approaches. In particular, the fusion of multimodal data substantially enhances the model's capacity to capture fine-grained features under complex traffic scenarios. HGNNs effectively model spatiotemporal dependencies, while the introduction of the CPLE algorithm further improves the model's sensitivity to subtle and emerging anomalies. Compared with existing methods, the proposed framework consistently maintains high performance even in challenging environments characterized by complexity and noisy situations, confirming its reliability and generalizability in real-world traffic applications.

Cross-Dataset generalization experiment

To comprehensively evaluate the generalization capability of the proposed framework, cross-dataset experiments were conducted. In these experiments, the model was trained on one dataset and tested on an unseen dataset to assess its adaptability to varying traffic conditions and environments. Specifically, the METR-LA dataset (Los Angeles traffic data) was used for training, while the PEMS-BAY dataset (San Francisco Bay Area traffic data) served as the testing set. This setup allowed for a realistic evaluation of the model's transferability across geographically and structurally distinct traffic networks. Evaluation was performed using five standard performance metrics: accuracy, precision, recall, F1 score, and AUC. The results of the cross-dataset generalization experiment are summarized in Table 8.

The experimental results indicate that the proposed MHGNN-CPLE model exhibits superior generalization performance across datasets. When evaluated on the unseen PEMS-BAY dataset, the model achieves an accuracy of 0.945 and an F1 score of 0.940, significantly outperforming the baseline models AGC-LSTM and AttentionDeepST. These results highlight the framework's strong adaptability to new traffic environments without the need for extensive retraining. The AGC-LSTM model also demonstrates reasonable generalization capability, but its performance is consistently lower than that of MHGNN-CPLE, with an accuracy of 0.920 and an F1 score of 0.915, respectively. In contrast, AttentionDeepST yields the weakest performance among the three models, attaining an accuracy of 0.905 and an F1 score of 0.895. This suggests that AttentionDeepST may struggle to capture the underlying patterns and structural variations in previously unseen traffic datasets. Overall, the cross-dataset generalization experiment underscores the robustness, scalability, and practical applicability of the MHGNN-CPLE framework. Its strong transferability across geographically distinct datasets makes it a promising solution for real-world highway traffic anomaly detection and prediction tasks, where

heterogeneous and dynamic traffic conditions are the norm. This generalization capability is essential for the deployment of reliable, adaptive, and region-independent intelligent traffic management systems.

Model robustness testing experiment

To further evaluate the robustness of the proposed model under varying noise conditions, a dedicated robustness testing experiment was conducted. In real-world traffic anomaly detection scenarios, input data is often affected by diverse forms of noise, such as sensor failures, environmental disturbances, or transmission errors. Therefore, ensuring the model's stability in noisy environments is essential for practical deployment.

In this experiment, Gaussian noise of varying intensities was added to the original dataset to simulate data corruption scenarios that may arise in real-world applications. The noise levels were set at 5%, 10%, 15%, and 20%. By evaluating model performance under these different noise conditions, we assessed each model's ability to maintain detection accuracy and reliability.

The experiment was conducted using the PEMS-BAY dataset, and the standard evaluation metrics were employed, including accuracy, precision, recall, F1 score, and AUC. The results of the robustness evaluation for each model are summarized in Table 9.

The experimental results show that as the noise intensity increases, the performance of all models decreases to some extent. However, the MHGNN-CPLE model demonstrates stronger noise resistance. At a noise intensity of 5%, the MHGNN-CPLE model achieves an accuracy of 0.960, an F1 score of 0.960, and an AUC of 0.980. Even at a high noise intensity of 20%, the MHGNN-CPLE model still maintains an accuracy of 0.945, an F1 score of 0.945, and an AUC of 0.965. In contrast, the performance of the AGC-LSTM and AttentionDeepST models declines more statistically significant. For instance, at a noise intensity of 20%, the AGC-LSTM model has an accuracy of 0.930 and an F1 score of 0.927, while the AttentionDeepST model has an accuracy of 0.920 and an F1 score of 0.917. This indicates that the MHGNN-CPLE model can perform stably across different noise levels, showing better robustness to data noise. It is therefore more suitable for the complex data environments encountered in practical traffic anomaly event detection.

Failure case analysis

Despite the high overall accuracy of the proposed framework, some failure cases were identified during validation. These misclassifications provide valuable insights into the limitations of the model.

One common failure occurred in scenarios involving adverse weather. For instance, heavy fog and light snow often reduced vehicle speeds and traffic flow, resulting in a misclassification as traffic congestion rather than weather-induced anomalies. In one case, where the actual anomaly was due to fog (ground truth probability 0.87), the model predicted congestion with a probability of 0.83. This highlights a challenge in distinguishing visually and behaviorally similar anomalies when relying solely on multimodal sensor data.

Another failure mode involved road maintenance activities, which generated audio and visual cues similar to those produced during minor accidents. In such instances, the model falsely detected accidents when the event was merely scheduled maintenance. For example, a road repair segment with moderate traffic slowdown and machinery noise was incorrectly labeled as an accident (predicted probability: 0.79; ground truth: 0.18).

These cases underscore the need for future enhancements in anomaly differentiation, possibly through integrating metadata (e.g., maintenance schedules) or vehicle-to-infrastructure communication.

Ablation studies

To assess the contribution of individual components and input modalities in the proposed model, a comprehensive ablation study was conducted. The ablation experiments were divided into two categories: module-level ablation and input-level ablation.

Module-level ablation

We evaluated the impact of three critical components in our model: the Hierarchical Graph Neural Network (HGNN), the attention mechanism, and the CPLE optimization algorithm. Each component was independently

Noise intensity	Model algorithm	Accuracy	Precision	Recall	F1 score	AUC
5%	MHGNN-CPLE	0.960	0.970	0.950	0.960	0.980
	AGC-LSTM	0.945	0.955	0.935	0.945	0.975
	AttentionDeepST	0.935	0.945	0.925	0.935	0.965
10%	MHGNN-CPLE	0.955	0.965	0.945	0.955	0.975
	AGC-LSTM	0.940	0.950	0.925	0.937	0.970
	AttentionDeepST	0.930	0.940	0.915	0.927	0.960
15%	MHGNN-CPLE	0.950	0.960	0.940	0.950	0.970
	AGC-LSTM	0.935	0.945	0.920	0.932	0.965
	AttentionDeepST	0.925	0.935	0.910	0.922	0.955
20%	MHGNN-CPLE	0.945	0.955	0.935	0.945	0.965
	AGC-LSTM	0.930	0.940	0.915	0.927	0.960
	AttentionDeepST	0.920	0.930	0.905	0.917	0.950

Table 9. Model robustness testing results.

Model variant	Accuracy	F1-score	Precision	Recall
Full model (HGNN + Attn + CPLE)	0.975	0.962	0.960	0.965
w/o HGNN (use GCN)	0.942	0.930	0.925	0.934
w/o attention mechanism	0.948	0.936	0.933	0.939
w/o CPLE (use Adam)	0.940	0.926	0.930	0.920

Table 10. Impact of model components on anomaly detection Performance.

Input modalities used	Accuracy	F1-score
All modalities	0.975	0.962
Image + point cloud only	0.951	0.937
w/o sound	0.965	0.940
w/o Temperature	0.958	0.942
Image only	0.930	0.914
Point cloud only	0.922	0.906

Table 11. Contribution of each sensor modality.

removed or replaced with a simpler baseline (e.g., standard GCN for HGNN, average pooling for attention, and Adam optimizer instead of CPLE). The results of the model robustness testing are presented in Table 10.

The results clearly indicate that each of these modules contributes significantly to the overall model performance. The removal of HGNN leads to a sharp decrease in spatial structure understanding. The attention mechanism improves robustness to redundant or irrelevant features, while CPLE enhances convergence and performance, particularly under noisy conditions.

Input-level ablation

We also conducted input modality ablation experiments by removing one or more sensor inputs (image, point cloud, sound, temperature) and retraining the model.

As shown in Table 11, the results demonstrate that image and point cloud data are the dominant modalities for visual and spatial analysis. However, the exclusion of sound and temperature data leads to a noticeable performance drop, particularly in detecting hidden or weather-related anomalies. This highlights the necessity of these modalities in improving generalization, especially in real-world environments with complex traffic and environmental conditions.

Diagnostic interpretation of ablation study

Each component of the proposed model contributes uniquely to its performance. Diagnostic analysis of the ablation results reveals the following insights:

- Removal of HGNN: Replacing HGNN with standard GCNs led to a 3.2% drop in accuracy and a 3.6% drop in F1-score. This is primarily due to HGNN's ability to capture heterogeneous relations between modalities. Without this capability, the model struggled to fuse spatially and semantically diverse data, especially in accident and congestion scenarios.
- Removal of Attention Mechanism: Excluding the attention mechanism caused the model to lose its dynamic weighting capability. In weather-related scenarios where the relevance of each modality fluctuates (e.g., temperature gaining prominence), the absence of attention resulted in poor discrimination of subtle anomalies.
- Removal of CPLE: Replacing CPLE with the Adam optimizer led to a decline in performance under noisy conditions. The CPLE algorithm improves robustness by dynamically adjusting sample weights based on prediction uncertainty. Without it, the model showed reduced recall in rare anomaly classes, such as road icing or early-stage congestion.

These findings emphasize the complementary roles of each component in enhancing both generalization and reliability.

Discussion and future work

The integration of multimodal deep fusion and HGNNs represents a significant advancement in the detection and prediction of highway traffic anomalies. By leveraging diverse sensor data—images, LiDAR, sound, and temperature—the framework successfully captures complex spatiotemporal dependencies, yielding higher accuracy and robustness compared to baseline models. The ensemble CPLE optimization further enhances reliability by adaptively weighting uncertain samples.

Nevertheless, several practical limitations must be acknowledged. First, the model depends on synchronized, high-resolution multimodal sensor input. In real-world deployments, sensors may fail or produce degraded outputs due to environmental noise or hardware faults. Missing or asynchronous data could reduce the system's performance. Second, the computational complexity of the HGNN and fusion modules poses challenges for real-

time inference, particularly on edge devices with limited processing power. Third, while our model demonstrates strong generalization on cross-validation datasets, all data originates from California highway scenarios. Its transferability to different geographic regions or sensor infrastructures remains untested.

To address these issues, future work will explore lightweight architectures through pruning, quantization, or knowledge distillation, enabling real-time deployment. We also plan to develop robust sensor dropout mechanisms to tolerate partial input failure. In addition, domain adaptation techniques and few-shot learning will be investigated to improve generalization across diverse environments. Finally, real-world pilot deployments in urban tunnels and highway intersections are planned to evaluate system latency, fault tolerance, and practical integration feasibility.

Conclusion

The framework for detecting and predicting abnormal traffic events on highways, which integrates multimodal deep fusion and heterogeneous graph neural networks (HGNNs), effectively consolidates multiple data sources. By harnessing the spatiotemporal modeling capabilities of graph neural networks alongside the optimization techniques of the CPLE algorithm, it substantially enhances the accuracy and robustness of detecting traffic anomalies. This approach has shown great potential in boosting detection performance. While current results are promising, their generalizability remains to be fully validated. The proposed model demonstrates consistent performance on METR-LA and PEMS-BAY datasets, but both datasets originate from limited geographic regions and share similar traffic sensor infrastructures. This raises the possibility that the model's performance may vary when deployed in different environmental or infrastructural conditions. While incorporating broader datasets is beyond the scope of this study, future work will expand validation efforts to include diverse urban and highway networks. In addition to generalizability, practical implementation still faces several challenges, such as data quality, inconsistencies, and the demand for real-time processing. Given the highly dynamic and time-sensitive nature of traffic conditions, it is crucial to overcome these limitations to achieve optimal performance. Future research should focus on several key directions: refining data fusion techniques to better handle diverse and incomplete data, boosting the computational efficiency of models to satisfy real-time processing needs, and delving into additional data sources, like sensor networks, social media data, and vehicle - to - vehicle communications, to gain a more all - encompassing view of traffic events. These advancements are expected to give rise to more precise, scalable, and intelligent traffic management systems, which in turn will greatly enhance road safety and traffic flow.

Moreover, several practical limitations of the current framework warrant further attention. First, the computational complexity introduced by components such as HGNNs and the CPLE optimization may hinder deployment on low-resource or real-time systems. Second, the reliance on multiple sensor modalities raises concerns about robustness in cases of sensor failure or incomplete data streams. Finally, while the model performs well in controlled experiments, its real-time applicability remains limited due to processing delays, particularly in graph construction and feature fusion. Addressing these issues in future work—through model simplification, robust modality dropout strategies, and real-time inference optimization—will be critical to ensuring practical scalability and deployment in intelligent traffic monitoring systems.

Data availability

The data are available from the corresponding author on reasonable request.

Received: 28 April 2025; Accepted: 3 September 2025
Published online: 29 September 2025

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Funding

This work was supported by the open project of the Key Laboratory of Traffic Information and Safety in Anhui Province's ordinary universities (JTX202302); Anhui Province Excellent Youth Project for Universities (2022AH030161); 3. Anhui Postdoctoral Scientific Research Program (No.2025B1094).

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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